

Quiet Neurons: a BDH quietens on what it just learned, not on what is merely predictable

DataForge 2026, Pathway track · concept **Sparse Non-Negative Activations** · Nilay Toshniwal, solo · artifact nilaymastaadmi.github.io/quiet-neurons · code github.com/nilaymastaadmi/quiet-neurons

THE DESIGN PRESSURE

A dense Transformer evaluates every feed-forward unit for every token, hard or not; activation sparsity is the counter-proposal. But Transformer sparsity is already emergent, arriving with training rather than by design [3]; it needs a hard rectifier, because only that yields exact zeros [4]; and *which* units fire is input-dependent, not merely how many [5]. Sparsity alone therefore distinguishes nothing. What distinguishes an architecture is what its sparsity is a function of.

WHAT BDH CHANGES

Dragon Hatchling (BDH) [1] makes it structural. In the BDH-GPU formulation Pathway open-sources, every layer multiplies two rectified (ReLU) vectors elementwise, and a product of two sparse non-negative vectors is sparser than either. *Active* here means a strictly non-zero entry in that product; a rectifier cannot emit a negative, so an active neuron is a positive amount of something rather than a signed coordinate, which is what lets it be read as a concept. Attention is the second change: query and key are the same tensor, so the score matrix is a causally-masked Gram matrix of neuron activations with **no softmax**. Positions whose neurons fired alike bind to each other.

THE CLAIM, AND THE CORRECTION IT NEEDED

Section 6.4 reports that "fewer neurons are active ... for more predictable input signals": at layer 2 of a 65,536-neuron model, 4.0–7.5% active while memorising a new word against about 2.5% while repeating it, a 1.6–3.0× ratio [1, Fig. 14]. We reproduced that protocol on three models trained here from the public code, all for the same 2,309 steps, and it survives a 32-fold shrink: **1.47×, 2.12× and 1.87×** at 2,048, 8,192 and 16,384 neurons, each averaged over 1,280 sequences with spread near 0.005. **In aggregate the paper is right: predictable text is quieter.** What follows sharpens that rather than contradicting it.

The same measurement narrows what the result can mean. Two blocks of one sequence are both predicted almost perfectly yet differ 2.65-fold in activity: the 13-letter warm-up runs at **9.7%** of layer 2 active at 0.0004 nats of mean surprise (a nat is the natural-log unit of cross-entropy; zero is a perfect prediction), the 56 repeated letters at **3.6%** at 0.004 nats, and 4.6× apart at the extremes. The warm-up is identical in every training sequence, so it lives in the **weights**; the repeated word is new each sequence, so it can only come from the **context**. What separates them is not predictability but **where the knowledge came from**. We show that activity tracks that; we do not show why. A learner falsifies it in a minute: inject an unpredictable letter mid-repetition and activity rises there, while the letter before is unchanged to nine decimals, because the model reads only leftwards.

System	Where its sparsity comes from	Carried per token	Evidence status
Dense Transformer feed-forward	Emergent in training [3]; needs a hard rectifier for exact zeros [4]	A key-value cache, growing with context	Extensive, independent
BDH-GPU , measured here	Structural: elementwise product of two rectified vectors, sparser than either	A fixed-size synaptic state	Reproduced here at toy scale; author-reported at 65,536 neurons
BDH-CQ	Not public	Not public	Developer-reported only

EVIDENCE, LABELLED

REPRODUCED HERE: the three ratios, the counterexample, the surprise-injection check, and parity of the in-browser model against PyTorch (logit max difference 6.1e-5, all twelve sparsity series bit-identical). **DEVELOPER-REPORTED, NOT REPRODUCED:** the Figure 14 band; 97.4% on Sudoku Extreme, which Pathway's own repository says comes from their internal implementation and does not reproduce from the open code; the 1B-to-600B scaling experiments; and BDH-CQ's 29.5% pass@2 on ARC-AGI-1 at \$0.0007 per task [2]. **NOT MODELLED:** BDH-CQ's internals, "proprietary" by its authors' own statement [2]; **it has no direct role in this concept**, and inventing one would be worse than saying so. The strongest external grounding Pathway offers is development on Amazon SageMaker HyperPod, a commercial partnership rather than an independent evaluation. **WHERE IT IS LIKELY WORSE:** sparse non-negative activations plausibly cost capacity against dense superposition, and no published comparison runs past GPT-2 scale.

LIMITATIONS, MATURITY, AND WHAT WOULD SETTLE THEM

The effect is not uniform with depth: layer 0 runs *backwards* at all three sizes (0.80, 0.80, 0.86) and layer 3 is flat below 16,384 neurons, weakly positive (1.11×) there. Scaling is **not monotonic**, and we published the opposite before the third model finished. The obvious confound is gone: the three were originally cut by wall clock at 1,854, 1,917 and 2,309 steps of one schedule, so we retrained the short two to 2,309. Both rose, 1.43 to 1.47 and 1.97 to 2.12, and the dip at 16,384 survived, now forty times the sampling spread. These are toys, 0.4M to 6.3M parameters on a synthetic task, as the paper's protocol is. Surprise and sparsity do *not* track per letter (8,192 neurons, layer 2: Pearson 0.30, Spearman -0.05), which killed a stronger claim. We count the elementwise product, the only quantity in the public code reaching the paper's band, and the paper does not say which tensor it means; that changes the effect's size, not its direction. Most importantly we establish the *signature*, not its cause: nothing explains why context-held knowledge needs fewer neurons, and the control that would settle it relocates identical content between weights and context. **MATURITY:** credible and outsider-checkable for this narrow property; the wider programme is earlier, its headline benchmarks developer-reported.

[1] Kosowski, Uznański, Chorowski, Stamirowska, Bartoszkiewicz, *The Dragon Hatchling*, arXiv:2509.26507 (2025), §6.4 and Fig. 14. [2] Engdahl, Kosowski, Chorowski et al., *BDH-CQ*, arXiv:2608.09888 (2026). [3] Li, You, Bhojanapalli et al., *The Lazy Neuron Phenomenon*, arXiv:2210.06313 (2022), ICLR 2023. [4] Mirzadeh, Alizadeh, Mehta et al., *ReLU Strikes Back*, arXiv:2310.04564 (2023). [5] Liu, Wang, Dao et al., *Deja Vu*, arXiv:2310.17157 (2023), ICML 2023. — Reproduce every number: `python measure.py --ckpt checkpoints/bdh_n2048.pt --embd 64 --mult 32 --repeats 5`. Code and data MIT; bdh.py is Pathway's, unmodified, MIT.